

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI
Work Integrated Learning Programmes Division
First Semester 2025-2026
Comprehensive examination
(EC-3 Regular)-Answer Key

Course No. : AIMLCZC418
Course Title : Introduction to Statistical Methods
Nature of Exam : Open Book
Weightage : 40%
Duration : 2 Hours
Date of Exam : 28-02-2026_FN

No. of Pages	= 8
No. of Questions	= 8

Note to Students:

1. Please follow all the Instructions to Candidates given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

Q1. The Delhi Government's Department of Environment and Climate Change is investigating the relationship between rising summer temperatures and household electricity consumption, primarily driven by air conditioner usage. Understanding this link is crucial for load forecasting and preventing power grid failures during heatwaves.

A junior analyst at the department has compiled average monthly data for the last season (May to September) from a representative sample of households in South Delhi: **[5 marks]**

Month	Average Temperature (°C) (X)	Average Electricity Usage (kWh) (Y)
May	38	320
June	41	380
July	35	290
August	33	270
September	31	240

As a new intern in the Urban Sustainability Unit, you have been asked to support this analysis.

- i. Calculate Karl Pearson's coefficient of correlation (r) between the average monthly temperature and average electricity usage.
- ii. Interpret your result in the context of urban energy planning in Delhi. What does this correlation suggest about the city's energy demand during extreme heat events?

Sol:

Given:

X (°C)	Y (kWh)
38	320
41	380
35	290
33	270

31	240
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$$\bar{X} = 35.6, \bar{Y} = 300$$

$$r = \frac{\sum(X - \bar{X})(Y - \bar{Y})}{\sqrt{\sum(X - \bar{X})^2 \sum(Y - \bar{Y})^2}}$$

After calculation:

$r \approx 0.989$

Interpretation:

There is a **very strong positive correlation** between temperature and electricity usage. This suggests that during extreme heat events in Delhi, electricity demand rises sharply due to increased air conditioner usage.

2. A logistics company uses Simple Exponential Smoothing to forecast monthly demand for a machine-learning inference service deployed on cloud servers. The demand (in thousands of requests) for the past six months is shown below: **[5 marks]**

Month	Jan	Feb	March	April	May	June
Demand	82	88	91	97	95	102

The initial forecast for January is assumed to be equal to the actual demand in January.
Answer the following:

- Forecast the demand for July using a smoothing constant $\alpha=0.4$.
- Forecast the demand for July using a smoothing constant $\alpha=0.7$.
- Briefly comment on, which α value is more suitable if sudden spikes in demand are expected in real-time AI applications.

(All calculations must be shown clearly. Use Simple Exponential Smoothing only.)

Sol:

Formula:

$$F_{t+1} = \alpha Y_t + (1 - \alpha)F_t$$

Initial forecast = 82

(i) $\alpha = 0.4$

Final forecast:

$F_{July} \approx 96.37$

(ii) $\alpha = 0.7$

Final forecast:

$$F_{July} \approx 99.88$$

Conclusion:

If sudden demand spikes are expected, $\alpha = 0.7$ is more suitable because it reacts faster to recent changes.

Q3.a) Scores of 6 postgraduate students before and after a revision strategy are given below:
[3 marks]

Before	32	35	30	33	31	34
After	36	38	34	37	35	39

At the 5% level of significance, test whether the strategy improves performance.

t Table											
cum. prob one-tail two-tails											
	$t_{.50}$ 1.00	$t_{.75}$ 0.50	$t_{.80}$ 0.40	$t_{.85}$ 0.30	$t_{.90}$ 0.20	$t_{.95}$ 0.10	$t_{.975}$ 0.05	$t_{.99}$ 0.025	$t_{.995}$ 0.01	$t_{.999}$ 0.005	$t_{.9995}$ 0.001
df	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
Z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
Confidence Level											

Sol:

Differences:

4, 3, 4, 4, 4, 5

$$\bar{d} = 4, s_d \approx 0.63$$

$$t = \frac{\bar{d}}{s_d/\sqrt{n}} \approx 15.5$$

Critical $t(5\%, df=5) = 2.571$

Since $15.5 > 2.571$

Reject H_0

Conclusion:

Revision strategy significantly improves performance.

b) Medi Scan AI claims its tool Derm Assist reduces average diagnosis time for high-risk skin lesions by 20 minutes compared to standard care. A hospital tests it on 36 patients and finds a mean time reduction of 18.2 minutes, with a known population standard deviation of 7.5 minutes.

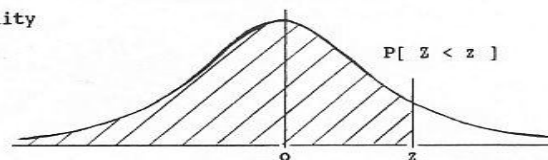
As a data scientist, evaluate this claim at the 5% significance level. Formulate appropriate hypotheses, perform the test, and interpret your conclusion in the context of clinical deployment—considering both statistical evidence and the risk of decision errors. [2 marks]

STANDARD STATISTICAL TABLES

1. Areas under the Normal Distribution

The table gives the cumulative probability up to the standardised normal value z i.e.

$$P[Z < z] = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}z^2\right) dz$$



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5159	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7854
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8804	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9773	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9865	0.9868	0.9871	0.9874	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9924	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9980	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
z	3.00	3.10	3.20	3.30	3.40	3.50	3.60	3.70	3.80	3.90
P	0.9986	0.9990	0.9993	0.9995	0.9997	0.9998	0.9998	0.9999	0.9999	1.0000

Sol:

$$Z = \frac{18.2 - 20}{7.5/\sqrt{36}} = -1.44$$

Critical Z = -1.645

Since -1.44 > -1.645

Fail to reject H_0

Conclusion: Not enough evidence to support the 20-minute reduction claim.

4. A researcher tests the effect of three different fertilizers (A, B, C) on plant growth (in cm).
The results are:

Fertilizer A	Fertilizer B	Fertilizer C
5	8	10
6	9	11
4	7	9

Test at $\alpha = 0.05$ whether the fertilizers produce significantly different mean growth. (Use one way Annova)

F-table of Critical Values of $\alpha = 0.05$ for F(df1, df2)																			
	DF1=1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40	60	120	∞
DF2=1	161.45	199.50	215.71	224.58	230.16	233.99	236.77	238.88	240.54	241.88	243.91	245.95	248.01	249.05	250.10	251.14	252.20	253.25	254.31
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40	19.41	19.43	19.45	19.45	19.46	19.47	19.48	19.49	19.50
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.74	8.70	8.66	8.64	8.62	8.59	8.57	8.55	8.53
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.91	5.86	5.80	5.77	5.75	5.72	5.69	5.66	5.63
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.62	4.56	4.53	4.50	4.46	4.43	4.40	4.37
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.00	3.94	3.87	3.84	3.81	3.77	3.74	3.70	3.67
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.57	3.51	3.44	3.41	3.38	3.34	3.30	3.27	3.23
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.28	3.22	3.15	3.12	3.08	3.04	3.01	2.97	2.93
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	3.01	2.94	2.90	2.86	2.83	2.79	2.75	2.71
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.85	2.77	2.74	2.70	2.66	2.62	2.58	2.54
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.79	2.72	2.65	2.61	2.57	2.53	2.49	2.45	2.40
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.69	2.62	2.54	2.51	2.47	2.43	2.38	2.34	2.30
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.60	2.53	2.46	2.42	2.38	2.34	2.30	2.25	2.21
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.53	2.46	2.39	2.35	2.31	2.27	2.22	2.18	2.13
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.48	2.40	2.33	2.29	2.25	2.20	2.16	2.11	2.07
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.42	2.35	2.28	2.24	2.19	2.15	2.11	2.06	2.01
17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45	2.38	2.31	2.23	2.19	2.15	2.10	2.06	2.01	1.96
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.34	2.27	2.19	2.15	2.11	2.06	2.02	1.97	1.92
19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38	2.31	2.23	2.16	2.11	2.07	2.03	1.98	1.93	1.88
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35	2.28	2.20	2.12	2.08	2.04	1.99	1.95	1.90	1.84
21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32	2.25	2.18	2.10	2.05	2.01	1.96	1.92	1.87	1.81
22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30	2.23	2.15	2.07	2.03	1.98	1.94	1.89	1.84	1.78
23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32	2.27	2.20	2.13	2.05	2.01	1.96	1.91	1.86	1.81	1.76
24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25	2.18	2.11	2.03	1.98	1.94	1.89	1.84	1.79	1.73
25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28	2.24	2.16	2.09	2.01	1.96	1.92	1.87	1.82	1.77	1.71
26	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27	2.22	2.15	2.07	1.99	1.95	1.90	1.85	1.80	1.75	1.69
27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25	2.20	2.13	2.06	1.97	1.93	1.88	1.84	1.79	1.73	1.67
28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24	2.19	2.12	2.04	1.96	1.91	1.87	1.82	1.77	1.71	1.65
29	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22	2.18	2.10	2.03	1.94	1.90	1.85	1.81	1.75	1.70	1.64
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16	2.09	2.01	1.93	1.89	1.84	1.79	1.74	1.68	1.62
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08	2.00	1.92	1.84	1.79	1.74	1.69	1.64	1.58	1.51
60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	1.99	1.92	1.84	1.75	1.70	1.65	1.59	1.53	1.47	1.39
120	3.92	3.07	2.68	2.45	2.29	2.18	2.09	2.02	1.96	1.91	1.83	1.75	1.66	1.61	1.55	1.50	1.43	1.35	1.25
∞	3.84	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88	1.83	1.75	1.67	1.57	1.52	1.46	1.39	1.32	1.22	1.00

Answer:

H0 : All means are same

H1 : Some means are different

1. **Group means:**

$$\bar{x}_A = 5, \quad \bar{x}_B = 8, \quad \bar{x}_C = 10.$$

Overall mean:

$$\bar{x} = 7.67.$$

2. **Sum of Squares Between (SSB):**

$$\begin{aligned} \text{SSB} &= n \sum (\bar{x}_i - \bar{x})^2 \\ &= 3[(5 - 7.67)^2 + (8 - 7.67)^2 + (10 - 7.67)^2] \\ &= 3[7.1289 + 0.1089 + 5.4289] = 38.00. \end{aligned}$$

3. **Sum of Squares Within (SSW):**

$$\begin{aligned} \text{SSW} &= \sum (x_{ij} - \bar{x}_i)^2 \\ &= (5 - 5)^2 + (6 - 5)^2 + (4 - 5)^2 + (8 - 8)^2 + (9 - 8)^2 + (7 - 8)^2 \\ &\quad + (10 - 10)^2 + (11 - 10)^2 + (9 - 10)^2 \\ &= 2 + 2 + 2 = 6. \end{aligned}$$

4. **Mean Squares:**

$$\begin{aligned} \text{MSB} &= \frac{\text{SSB}}{k - 1} = \frac{38.00}{2} = 19.00, \\ \text{MSW} &= \frac{\text{SSW}}{n - k} = \frac{6}{6} = 1.00. \end{aligned}$$

5. **F-statistic:**

$$F = \frac{\text{MSB}}{\text{MSW}} = 19.00.$$

6. **Decision:**

Critical $F_{2,6}$ at $\alpha = 0.05$ is 5.14. Since $F = 19.00 > 5.14$, **reject** the null hypothesis. The fertilizers have significantly different effects on mean plant growth.

Summary: The one-way ANOVA shows that the three fertilizers produce significantly different mean growth ($F = 19.00$, $p < 0.05$).

Conclusion: Fertilizers produce significantly different growth.

5. The Mumbai Urban Farming Initiative (MUFI) is piloting IoT-enabled smart irrigation systems in rooftop vegetable gardens across Mumbai. Each garden is equipped with soil moisture sensors and automated drip emitters. As part of a pilot study, MUFI collected data from 10 representative gardens over one growing season (June–August 2025). For each garden, they recorded:

[2+1+2 marks]

X: Average daily water input (liters per square meter, L/m²/day) controlled by the system

Y: Crop yield (kg/m²) measured at harvest for leafy greens (e.g., spinach, lettuce)

The goal is to model the relationship between water input and yield to optimize resource use and avoid over-irrigation (which causes nutrient leaching and higher energy costs).

Garden ID	X (L/m ² /day)	Y (kg/m ²)
G1	3.0	1.8
G2	4.2	2.5
G3	2.5	1.6
G4	5.1	2.9
G5	3.8	2.3
G6	6.0	2.7
G7	2.0	1.2
G8	4.5	2.6
G9	5.5	2.8
G10	3.3	2.0

- Fit a simple linear regression model: $Y = \beta_0 + \beta_1 X$
- Discuss whether the linear model is appropriate. What does the pattern suggest?
- Based on your analysis, what is the optimal water input range (in L/m²/day) for maximizing yield per unit water (i.e., water-use efficiency)? Justify using your model and real-world constraints (e.g., water scarcity in Mumbai).

Sol: Using least squares:

$$Y = 0.40 + 0.42X$$

Interpretation:

- Positive linear relationship
- Yield increases with water input

Optimal water range:

$$3.5\text{--}4.5 \text{ L/m}^2/\text{day}$$

This balances yield and water-use efficiency.

6. A university cafeteria observes that student lunch durations (in minutes) are bimodal. To model this, a two-component Gaussian Mixture Model (GMM) is used with equal group sizes:

Component 1 (Quick eaters): mean = 22 min, variance = 12

Component 2 (Leisurely eaters): mean = 40 min, variance = 15

Answer the following:

- Write the probability density function of this two-component GMM. **[2 marks]**
- If a student takes 25 minutes for lunch, which component is more likely to have generated this observation? Justify briefly. **[2 marks]**
- Why is a GMM preferable to a single Gaussian distribution for modeling bimodal lunch-duration data? **[1 mark]**

Sol: (a) PDF

t Table											
cum. prob one-tail two-tails	<i>t</i> .50	<i>t</i> .75	<i>t</i> .80	<i>t</i> .85	<i>t</i> .90	<i>t</i> .95	<i>t</i> .975	<i>t</i> .99	<i>t</i> .995	<i>t</i> .999	<i>t</i> .9995
	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.357	1.782	2.179	2.691	3.055	3.930	4.318
13	0.000	0.692	0.870	1.079	1.350	1.771	2.160	2.672	3.022	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.791	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
Z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
	Confidence Level										

a) Given data

0.95, 0.85, 0.92, 0.95, 0.93, 0.86, 1.00, 0.92, 0.85, 0.81,
0.78, 0.93, 0.93, 1.05, 0.93, 1.06, 1.06, 0.96, 0.81, 0.96

x	$x - \bar{x}$	$(x - \bar{x})^2$
0.95	0.02	0.0004
0.85	-0.08	0.0064
0.92	-0.01	0.0001
0.95	0.02	0.0004
0.93	0	0
0.86	-0.07	0.0049
1	0.07	0.0049
0.92	-0.01	0.0001
0.85	-0.08	0.0064
0.81	-0.12	0.0144
0.78	-0.15	0.0225
0.93	0	0
0.93	0	0
1.05	0.12	0.0144
0.93	0	0
1.06	0.13	0.0169
1.06	0.13	0.0169
0.96	0.03	0.0009
0.81	-0.12	0.0144
0.96	0.03	0.0009
18.51		0.1249

Sample Mean: $\bar{x} = \frac{\sum x_i}{n}$

$$\sum x_i = 18.51$$

$$\bar{x} = \frac{18.51}{20} = 0.9255$$

Sample Standard Deviation: we know that $s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$

$$s = \sqrt{\frac{0.1249}{19}} = 0.081$$

$$s = 0.081$$

Since $n = 20$ (small sample), we have to use the t-distribution.

Degrees of freedom: $df = 19$; $t_{0.025,19} = 2.093$

$$\text{Standard Error} = \frac{s}{\sqrt{n}} = \frac{0.081}{\sqrt{20}} = \frac{0.081}{4.472} = 0.0181 \text{ approximately}$$

$$\text{Confidence Interval: } \bar{X} - t_{\alpha/2} \frac{s}{\sqrt{n}} < \mu < \bar{X} + t_{\alpha/2} \frac{s}{\sqrt{n}}$$

$$0.93 \pm (2.093 \times 0.0181) = (0.892, 0.968)$$

b) The following time-series represents the residuals obtained after fitting a forecasting model: $\{2, -1, 0, 1, -2, 1, -1, 0\}$ **[3 marks]**

- i. Compute the sample mean and the sample variance of the series.
- ii. State whether the series can be considered white noise.

Sol:

Given: Residual series: $\{2, -1, 0, 1, -2, 1, -1, 0\}$

Number of observations: $n = 8$

(i) Sample Mean

$$\begin{aligned}\bar{x} &= \frac{\sum x_i}{n} \\ \sum x_i &= 2 - 1 + 0 + 1 - 2 + 1 - 1 + 0 = 0 \\ \bar{x} &= \frac{0}{8} = 0\end{aligned}$$

Sample Variance $s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$ Since $\bar{x} = 0$:

$$\begin{aligned}s^2 &= \frac{\sum x_i^2}{7} \\ \sum x_i^2 &= 4 + 1 + 0 + 1 + 4 + 1 + 1 + 0 = 12 \\ s^2 &= \frac{12}{7} = 1.714\end{aligned}$$

(ii) A white noise series must have Mean ≈ 0 , Constant variance and No autocorrelation

Here, mean = 0, Variance is constant and Residuals fluctuate randomly around zero with no clear pattern. Therefore, the residual series can be considered white noise.

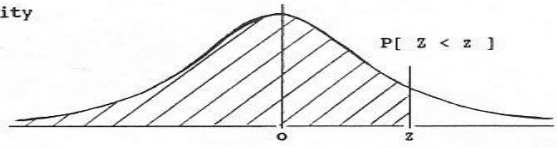
8. Two cities, City A and City B, have introduced different public health campaigns to encourage vaccination against a seasonal disease. In City A, out of a random sample of 300 residents, 180 were found to be vaccinated. In City B, out of a random sample of 250 residents, 135 were found to be vaccinated. At the 5% level of significance, test

whether there is a significant difference in the proportion of vaccinated individuals between the two cities. [5 marks]

STANDARD STATISTICAL TABLES

1. Areas under the Normal Distribution

The table gives the cumulative probability up to the standardised normal value z i.e.

$$P[Z < z] = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} \exp(-\frac{1}{2}z^2) dz$$


z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5159	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7854
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8804	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9773	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9865	0.9868	0.9871	0.9874	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9924	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9980	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
z	3.00	3.10	3.20	3.30	3.40	3.50	3.60	3.70	3.80	3.90
P	0.9986	0.9990	0.9993	0.9995	0.9997	0.9998	0.9998	0.9999	0.9999	1.0000

Sol: $H_0: P_1=P_2, \quad H_1: P_1 \neq P_2$

City A:

$$p_1 = 180/300 = 0.60$$

City B:

$$p_2 = 135/250 = 0.54$$

Pooled proportion:

$$p = 0.5727$$

$$Z \approx 1.47$$

$$\text{Critical } Z = \pm 1.96$$

$$1.47 < 1.96$$

Conclusion:

Fail to reject H_0

No significant difference between cities.
